

MUHAMMAD HUZAIFA

Oulu, Finland

+358 (46) 664 9442 / mhuzaifa404@proton.me

linkedin.com/in/huzaiifaaaa / huzaifa.eu

About

Full Stack Engineer specializing in payment infrastructure and security-critical systems, from PCI-DSS compliant gateways processing thousands of daily transactions to embedded cryptographic systems for defense infrastructure. 3+ years shipping production systems across FinTech and DefTech, proficient across Java (Spring Boot) and C#/.NET microservices, React, and cloud infrastructure (Azure, Kubernetes, CI/CD).

Technical Skills

- **Languages:** Java, C#, Python, Go, JavaScript, TypeScript, SQL, C, C++
- **Frameworks:** Spring Boot, ASP.NET Core, Node.js, React JS, REST APIs, API Design, Microservices, Hibernate
- **Databases:** PostgreSQL, MSSQL, MySQL, YugabyteDB
- **Messaging & Event-Driven Systems:** Apache Kafka, RabbitMQ
- **Cloud & DevOps:** Azure, Google Cloud, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Observability & Monitoring (Azure Sentinel)
- **Software Engineering:** System Design, Distributed Systems, OOP, Agile/Scrum, Unit Testing, Test Automation (Playwright, Selenium)
- **Security & Compliance:** Payment Card Industry Data Security Standard (PCI-DSS), GDPR, OWASP Top 10, SIEM
- **AI Tools:** Claude Code, Codex, Cursor, GitHub Copilot

Experience

Research Intern

June 2026 – August 2026

University of Oulu

Oulu, Finland

- Extended the AVISE security evaluation framework to Text-to-Image models, designing a 16-case adversarial test suite for style-escalation resistance.
- Co-authored a research paper formalizing security vulnerability methodologies for generative AI systems, submitted to NordSec 2026.
- Built OpenRouter based model connectors enabling standardized, reproducible security evaluation across 4+ T2I model providers spanning Google, X-AI, OpenAI, and open-source diffusion architectures.

Full Stack Software Engineer

February 2024 – March 2026

Datapulse Technologies (FinTech)

Ireland (Remote)

- Architected and delivered a replacement payment gateway platform from legacy infrastructure, now serving 1000+ merchants, cutting production debugging time and improving system maintainability.
- Engineered a 25% performance gain and reduced latency on a legacy whitelabel payment architecture.
- Integrated 3 global acquirers (E-Comprocessing, Transflo, Rapyd) and digital wallets (Apple Pay, Google Pay).
- Consolidated separate payment systems into a single unified gateway, eliminating dual-maintenance overhead and unifying transaction visibility across card-present and card-not-present flows.
- Built and scaled a fault-tolerant microservices architecture (Java/Spring Boot) processing 1000+ transactions, monitored via Azure Sentinel.
- Stepped in as substitute lead for a 4 person engineering team, coordinating sprint planning and code reviews to keep the gateway rebuild on schedule.
- Owned CI/CD pipeline design (GitHub → Jenkins → Docker → Kubernetes), enabling zero-downtime, single-push production releases.
- Led PCI-DSS compliance readiness end-to-end, closing 15+ audit findings ahead of certification via SIEM driven access controls and audit logging.

Trainee Engineer (Part Time)

July 2022 – May 2024

Rapidev (DefTech)

Islamabad, Pakistan

- Integrated 2 hardware sensors (detectors, jammers, spoofers) into the RapiGuard Anti-UAV system via GraphQL, enabling real-time aerial threat detection and neutralization for critical defense infrastructure deployments, with ongoing maintenance and UI/performance optimizations.

- Developed REST APIs for third-party system integration, expanding RapiGuard’s interoperability with external defense software platforms.
- Architected a Key Management Application (KMA) for encryption key lifecycles, implementing secure UART-based cryptographic key injection into radio hardware — deployed across military communications systems in the Gulf region.
- Built a Tactical Situational Awareness mobile app in Flutter (Android) with Google Maps integration, delivering real-time geospatial visualization and device location tracking for field operators.

Projects

Nafalytics

Founder & Full Stack Engineer

Stock Portfolio Tracker

- Built and launched a portfolio tracking platform from scratch in 3 months, growing to 300 active users tracking real-time P/L and trade activity.
- Developed real-time profit/loss tracking, historical performance analytics, automated payouts, and a notification engine.
- Directed full product lifecycle and cloud deployment (Java Spring Boot, React JS, PostgreSQL, Azure) with automated CI/CD via GitHub Actions. Live at <https://nafalytics.com>

Education

MSc Computer Science & Engineering

September 2025 – June 2027

University of Oulu

Oulu, Finland

BSc Computer Science

August 2020 – May 2024

National University of Sciences & Technology

Islamabad, Pakistan

- **Publication:** OptiGuard: Generalized, Attention-Driven & Explainable Glaucoma Classification
 - * Achieved state of the art results on G-1020 & SMDG datasets.
 - * Awarded 1st Best Project distinction.
 - * Project details: <https://linktr.ee/optiguard>

Certifications

Junior Penetration Tester

December 2025

TryHackMe

Course Work

- Hands-on experience with OWASP Top 10 vulnerabilities including SQL Injection (SQLi), Cross-Site Scripting (XSS), IDOR, and Command Injection.
- Skilled in Authentication Bypass, Server-Side Request Forgery (SSRF), and File Inclusion (LFI/RFI) exploitation and remediation.